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ON THE PATH INTEGRAL QUANTIZATION OF CHIRAL BOSONS

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We show that, in the covariant Lagrangian formalism, a proper treatment of the gauge degree of freedom in a model of chiral bosons proposed by Siegel uncovers the presence of a Jacobian (a "Wess-Zumino action"): the group of gauge transformations gets quantized and the anomaly is absorbed.

Due to its relevance in the construction of string theories, quantization of chiral bosons has recently received a lot of attention.[1–8]

In the Lagrangian formulation, the following action for a left-moving scalar in 2-dimensions[a] was proposed by Siegel:[1]

$$S = \int d^2x \left[\partial_- \phi \partial_+ \phi + \frac{\lambda}{2} (\partial_- \phi)^2 \right]. \tag{1}$$

The connection between this action and the one proposed by Floreanini and Jackiw[2] in their formulation of self-dual field quantization has been clarified.[8] It has also been shown[3] that the quantum theory constructed from action (1) (which has a gauge-invariance called Siegel invariance[1]) suffers from an anomaly. Modified actions leading to an anomaly-free quantum theory have been subsequently considered using Faddeev-Popov and BRST method.[3–4]

As it is well-known, the existence of anomalies require a careful analysis of the quantization procedure. In particular, if the Faddeev-Popov method is employed, factorization of the gauge degree of freedom does not work as in the nonanomalous case. For gauge theories or gravitational theories coupled to Weyl fermions it has been shown that the proper treatment of gauge degrees of freedom uncovers the presence

[a] We use Minkowski metric $(-1, 1)$ and $x_\pm = \frac{1}{\sqrt{2}}(x_0 \pm x_1)$

of a Wess-Zumino action[9-13] which as first suggested by Faddeev-Shatashvili,[14] cancels out the anomaly. In fact, these developments were inspired by Polyakov's work on quantization of strings[15] showing that a gauge degree of freedom, decoupled at the classical level, reappears after quantization.

Following these ideas, we discuss in this note the path-integral quantization of the theory defined by action (1). Using the Faddeev-Popov procedure, we point out the appearance of a Fujikawa Jacobian[16] arising from the usual change of variables that one has to perform within this approach.

When this Jacobian is taken into account, the resulting generating functional for λ becomes Siegel-invariant and hence the functional integral on λ is trivial. One then remains with a chiral scalar $\phi(x)$ and the $(-)$ component of a vector field, $\varepsilon^-(x)$ (which describes the gauge orbit) as relevant variables in the resulting quantum theory.

Although we shall follow the usual Faddeev-Popov procedure which starts from inserting an appropriate "resolution of unity" into the generating functional[17] we think that the alternative path-integral method defining a BRST invariant measure[18-22] should lead to the same answer, as it has been shown to be the case for theories that include gravity.[20]

We start from the generating functional defined as a path-integral over ϕ and λ:

$$Z = \int D\phi D\lambda \exp[iS[\phi;\lambda]] \tag{2}$$

with S given by (1). As we stated above, λ represents, at the classical level a gauge degree of freedom. Indeed, the equations of motion derived from (1) read:

$$\begin{gathered}\frac{1}{2}(\partial_-\phi)^2 = 0\\ 2\partial_+\partial_-\phi + \partial_+(\lambda\partial_-\phi) = 0\,.\end{gathered} \tag{3}$$

The first equation implies $\partial_-\phi = 0$ so that the second one is automatically satisfied. λ drops out of the equations of motion and corresponds to a gauge degree of freedom. The (infinitesimal) transformations associated to this gauge invariance are

$$\begin{gathered}\delta\phi = \phi^{\varepsilon^-} - \phi = \varepsilon^-\partial_-\phi\\ \delta\lambda = \lambda^{\varepsilon^-} - \lambda = -2\partial_+\varepsilon^- - \lambda\overleftrightarrow{\partial}_-\varepsilon^-\,.\end{gathered} \tag{4}$$

These transformations correspond to infinitesimal re-parametrizations $\delta x^- = \varepsilon^-(x)$, $\delta x^+ = 0$ accompanied by a Weyl transformation. $\lambda(x)$ transforms as the $(--)$ component of a tensor field and $\varepsilon^-(x)$ is the $(-)$ component of an (infinitesimal) vector. We see from (4) that $\lambda(x)$ can be gauged to 0 using $\delta\lambda = -2\partial_+\varepsilon^-$.

In order to handle this gauge invariance at the quantum level we consider a resolution of unity in the form

$$1 = \Delta_{FP}[\lambda] \int \delta(F[\lambda^{\varepsilon^-}]) D\varepsilon^- . \tag{5}$$

Here $F[\lambda] = 0$ is a gauge condition, $\varepsilon^-(x)$ is the field describing the gauge orbit corresponding to Siegel transformations and Δ_{FP} is the Faddeev-Popov determinant; as an example in the $\lambda = 0$ gauge it reads:

$$\Delta_{FP}[\lambda] = \det \frac{\delta\lambda}{\delta\varepsilon} = \det D_\Delta[\lambda] \tag{6}$$

with

$$D_\Delta = -(2\partial_+ + \lambda\overleftrightarrow{\partial}_-)\delta(x-y) . \tag{7}$$

Generally, the Faddeev-Popov procedure consists of the insertion of the resolution of unity into the generating functional so as to factorize the gauge degree of freedom. It amounts in the present case to pass from the integration over λ to the integration over the gauge group parameter ε^-. Let us call λ_0 the value of λ satisfying the selected gauge condition,

$$F[\lambda]|_{\lambda=\lambda_0} = 0 . \tag{8}$$

We can then define a generating functional Weff in the form:

$$\exp[W_{\text{eff}}[\lambda_0]] = \Delta_{FP}[\lambda_0] \int D\varepsilon^- D\phi \exp[iS[\phi;\lambda_0,\varepsilon^-]] . \tag{9}$$

Confront this definition with that in Ref. 4:

$$\exp[iW[\lambda_0] = \Delta_{FP}[\lambda_0] \int D\phi \exp[iS[\phi;\lambda_0]] . \tag{10}$$

Were the theory (Siegel) invariant at the quantum level, W_{eff} would be independent of λ, the ε-integration in (9) would trivially factorize, and definitions (9) and (10) would be identical.

We know however[3] that there is an anomaly which has to be taken into account. Then, in trying to eliminate the ε-dependence in (9), using

$$S[\phi;\lambda_0,\varepsilon] = S[\phi^{\varepsilon^-};\lambda_0,0] \tag{11}$$

and performing the change of variables

$$\phi \rightarrow \phi' = \phi^{\varepsilon^-}$$

one has to include a nontrivial Fujikawa Jacobian:[16]

$$D\phi = J_s[\lambda_0, \varepsilon^-] D\phi' \tag{12}$$

which can be defined through the (regularized) identity

$$\int D\phi \exp[iS[\phi; \lambda_0, \varepsilon^-] = \det{}^{-1/2} D[\lambda_0^{-\varepsilon^-}] = J_s[\lambda_0, \varepsilon^-] \det{}^{-1/2} D[\lambda_0] \tag{13}$$

with

$$D[\lambda] = (\partial_- \partial_+ + \partial_- \lambda \partial_+) \delta(x - y). \tag{14}$$

Using this Jacobian we have

$$\exp[iW_{\rm eff}[\lambda_0] = \Delta_{FP}[\lambda_0] \int D\phi D\varepsilon^- J_s[\lambda_0, \varepsilon^-] \exp[iS[\phi; \lambda_0]. \tag{15}$$

We see that due to the presence of the Jacobian the integration over the orbit does not trivially factorize.

The Faddeev-Popov determinant in (10) can be represented as usual in terms of a ghost path-integral:

$$\Delta_{FP}[\lambda_0] = \int Db^{++} Dc^- \exp\left[\int d^2x b^{++} D_\Delta[\lambda_0] c^-\right] \tag{16}$$

and then we have

$$\exp[iW_{\rm eff}[\lambda_0] = \int D\varepsilon^- D\phi Db^{++} Dc^- J_s[\phi, \lambda_0] \exp[i(S + S_{\rm ghost})]. \tag{17}$$

It is clear at this point that $W_{\rm eff}$ is gauge invariant:

$$W_{\rm eff}[\lambda_0] = W_{\rm eff}[\lambda_0{}^{\eta^-}] \tag{18}$$

i.e., $W_{\rm eff}$ is independent of λ (this is not the case for W as defined in (10)). Exactly as it happens for gauge theories with Weyl fermions,[9] the presence of the Jacobian and the ε^--integration also ensures in this case that the generating functional does not depend on λ.

Indeed, if we write (17) with $\lambda_0{}^{\eta^-}$ instead of λ_0, we have, after changing variables ($\phi \rightarrow \phi^{\eta^-}$):

$$\exp[iW_{\text{eff}}[\lambda_0^{\eta^-}]] = \int D\varepsilon^- D\phi Db^{++} Dc^- \exp[i(S[\phi, \lambda_0] + S_{\text{ghost}}[\lambda_0^{\eta^-}, b^{++}, c^-])]$$

$$\times J_s[\lambda_0^{\eta^-}, \varepsilon] J_s[\lambda_0, -\eta^-] . \tag{19}$$

Now, it is easy to see that J_s satisfies

$$J_s[\lambda_0^{\eta^-}, \varepsilon^-] J_s[\lambda_0, -\eta^-] = J_s[\lambda_0, -\varepsilon^- \cdot \eta^-] . \tag{20}$$

Moreover, in trying to eliminate the η^--dependence from the Faddeev-Popov determinant, (or equivalently from the ghost action) we know there is an anomaly[3] (which is -26 times that of the scalar field):

$$\Delta_{FP}[\lambda_0^{\eta^-}] = J_\Delta[\lambda_0, \eta^-] \Delta_{FP}[\lambda_0] . \tag{21}$$

We have defined, in analogy with Eq. (12) a Jacobian J_Δ which accounts for the gauge dependence of Δ_{FP}.

Now, in (5) $D\varepsilon^-$ is a measure over the group of re-parametrizations $x^- \to x^- + \varepsilon^-$, ($x^+ \to x^+$) but Siegel gauge-invariance also includes a Weyl transformation δ_W,

$$\delta_W \lambda = \lambda \partial_- \varepsilon^- .$$

One has then to take into account the change in $D\varepsilon^-$ under this transformation which necessarily has to cancel that coming from the Faddeev-Popov determinant, so that identity (5) remains valid.

We then have, using all these facts, identity (18).

Hence, we see that λ, classically decoupled, disappears also at the quantum level due to the invariance of W_{eff}. As relevant variables one remains with the chiral boson $\phi(x)$ and the $(-)$ vector component $\varepsilon^-(x)$.

From the anomalous behavior of W^4 in Eq. (10):

$$\delta_\eta W[\lambda] = \frac{(26-1)}{48\pi} \int d^2x \lambda \partial_-^3 \eta^- \tag{22}$$

(at the one-loop level, with the 26 coming from the ghost contribution and the (-1) from the scalar) we can infer the behavior of the other term contributing to W_{eff}:

$$\delta_\eta W_J[\lambda] = -\frac{25}{48\pi} \int d^2x \lambda \partial_- \eta^- \tag{23}$$

with

$$W_J[\lambda] = W_{\text{eff}}[\lambda] - W[\lambda] . \tag{24}$$

One can also establish the connection between $W_{\text{eff}}[\lambda]$ and the modified action defined in Ref. 3 to get a nonanomalous theory. Indeed, Imbimbo and Schwimer[3] added to $S[\phi, \lambda]$ a noninvariant term so that the resulting model is anomaly-free:

$$\exp[i\tilde{W}[\lambda]] = \int D\phi Db^{++} Dc^{-} \exp[i(S + S_{\text{ghost}} + \tilde{S})] \tag{25}$$

with

$$\tilde{S} = \alpha \int \lambda \partial_{-}^{2} \phi d^2 x \tag{26}$$

and $\alpha = \dfrac{25}{48\pi}$ so that

$$\delta_\eta \tilde{W}[\lambda] = 0. \tag{27}$$

One has

$$\exp[i\tilde{W}[\lambda]] = \exp\left[i\alpha^2 \int \partial_{-}^{2} \lambda(x) K(x,y) \partial_{-}^{2} \lambda(y) d^2x d^2y \right] \det D[\lambda] \Delta_{FP} \tag{28}$$

with K such that

$$D[\lambda] K(x,y) = \delta(x-y). \tag{29}$$

Then $W_J[\lambda]$ can be identified with the exponential factor in (28), at least at the one-loop level. Note however that the modified Siegel action proposed in Ref. 3 cannot be coupled to gravity in a straightforward manner[5] while W_{eff} does not present this problem.

We see that in our derivation, W_J is not added by hand but it naturally appeared by following the usual steps in the Faddeev-Popov procedure, exactly as it happens for gauge and gravitational theories with Weyl fermions.[9–14]

In this context, the alternative proposal of Ref. 6 of considering a set of d chiral scalars and then adjusting $d = 26$ (precisely the string no-ghost condition) can be understood by noting that each one of the scalars contribute with the same Jacobian J_s so that in this case, instead of (23) one has

$$\delta_\eta W_J = \frac{-(26-d)}{48\pi} \int d^2 x \lambda \partial_{-}^{3} \eta . \tag{30}$$

One then has $\delta_\eta W_J = 0$ for $d = 26$ and $J = 1$;[a] the ε^{-}-integration factors out trivially and the resulting (nonanomalous) theory consists only of the original chiral scalars.

[a] The full Jacobian $J[\lambda, \eta]$ can be computed from the infinitesimal one $J[\lambda, \delta\eta]$ by iteration (see for example Ref. 23), and $\log J[\lambda, \delta\eta] = i\delta_\eta W_J$.

In summary, we have shown that, as it happens in other models where classical invariances become anomalous after quantization, also in the case of chiral bosons a careful application of the Faddeev-Popov procedure leads to a consistent quantum theory. Without addition of an *ad-hoc* term to the original action, the proper treatment of gauge degrees of freedom uncovers the presence of a Wess-Zumino term with the group of transformations as dynamical variable. Many questions remain open in this respect. In particular, the obtainability of these results using BRST invariance following the method developed in Refs. 18–22, and also the alternative treatment along the lines of Refs. 14 and 24. We hope to address to these questions in future works.

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